

Literature Review

Riley Webster

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1 Introduction

In astronomy, the group of objects that make up temporary phenomena such as supernovae (SNe), gamma ray bursts and tidal disruption events are known as transients. They are one of the most important phenomena to observe, as the data can lead to extraordinary discoveries such as the accelerating expansion of the Universe (Perlmutter et al., 1999; Riess et al., 1998). SNe are particularly important due to a specific type of supernova (SN) called type Ia Supernovae (SNe Ia) which are classified spectroscopically by the presence of a single ionised silicon line in their spectra near the peak luminosity (Ruiter & Seitenzahl, 2024).

When SNe Ia explode and are observed by a telescope, their luminosity can be plotted as a function of time. Following standardisation of the light curve, the light curves all reach a well known peak luminosity with some remaining scatter (Ginolin et al., 2025; Kelsey et al., 2022; Martin et al., 2024). This peak luminosity can be used to then find the distance to the location of the detonation, using these distance calculations the parameters that describe the properties of the Universe (Ω_m, H_0, w) can be determined.

However the current methods of standardisation show that there remains systematic errors in the observed and modelled luminosities of these SNe. If these systematic errors can be accounted for, then truly representative luminosities of these SNe can result in more precise cosmological calculations. One of the methods to find new factors to account for in standardisation, is the study of host galaxy properties and their correlation with SNe Ia populations. If the light curves can be standardised based on factors that are intrinsic to the SNe Ia progenitor then the more representative light curves can be obtained resulting in better determinations of Ω_m, H_0, w.

2 Supernova Ia physics and Progenitor Channels

2.1 Standardisation

Type Ia Supernovae (SNe Ia) are standardisable candles, which means that upon observation of an SN Ia, the light curve exhibits predictable behaviour(Martin et al., 2024; Rigault et al., 2020). With corrections for the stretch and colour of the SN the light curve can be

standardised to correct the observed light curve to find the intrinsic luminosity of the SNe Ia. The corrections for the light curves can be found in the Tripp equation 1 that accounts for the colour and stretch of the supernovae to determine the distance modulus (Tripp, 1998). Using this distance modulus you can then derive distances to the Supernovae.

$$\mu_{obs} = m_B^* - M_B + \alpha x_1 - \beta c \tag{1}$$

In equation 1 the distance modulus, μ_{obs}, is the difference between the apparent magnitude of the supernova and the absolute magnitude of the supernova. The remaining parameters are the stretch x_1 and the colour c where α and β are the stretch and colour coefficients where α falls between 0.14 and 0.15 and β is between 2.9 and 3.5 (Brout et al., 2022; Martin et al., 2024). The stretch parameter is the parameter that corrects for the width of the light curve due to different distributions and masses of ^{56}Ni in the ejecta of the supernova (Deckers et al., 2022). The colour is corrected due to the intrinsic components of the supernova and the surrounding environment such as dust, which can cause red shift of the light, altering the shape of the light curve (Kelsey et al., 2022). m_B^* is the apparent magnitude of the SNe Ia which is how bright the object appears from Earth and M_B is the absolute magnitude of the SNe Ia which is how bright the SNe Ia is from 10 parsecs away.

Using the resulting μ_{obs} from equation 1, you can acquire the luminosity distance d_L to the object using equation 2, noting that the coefficient 10 is in units of parsecs (Brout et al., 2022).

$$d_L = 10(10^{\mu/5}) \tag{2}$$

From this equation the luminosity distance can then be used to calculate cosmological parameters such as Universe density parameters for matter and radiation Ω_m and Ω_r and equation of state parameter which describes how the density of matter radiation and dark energy change in time w_m, w_r, w_Λ (Brout et al., 2022). Cosmological quantities are important, as they represent the contents of our universe and can be used to determine properties of the Universe. Thus, standardisation of the supernovae and accounting for the different stretches and colours that are observed in the light curves helps result in more accurate distance

calculations and better calculations of the cosmological parameters.

The standardisation of SNe Ia light curves are not perfect, Brout et al. (2022), Martin et al. (2024), Rigault et al. (2020), and Wiseman et al. (2023) all show how the Tripp equation still leaves residual error in the distance modulus determination, which can be seen in Figure 1. Thus research continues to determine how to improve standardisation methods that result in cosmological parameters that better represent the observed properties of the Universe. The Hubble residual is the measure of scatter in the calculation of the distance modulus between the observed distance modulus with the corrections made in equation 1, and the distance modulus predicted by a specific cosmology, μ_{cos} seen in equation 3. It has been found that the scatter after corrections using the Tripp equation is reduced to 0.15 magnitude, leading to the conclusion that further or different corrections are needed to standardise these light curves (Brout et al., 2022; Kelsey et al., 2022; Martin et al., 2024).

$$\Delta\mu = \mu_{obs} - \mu_{cos} \tag{3}$$

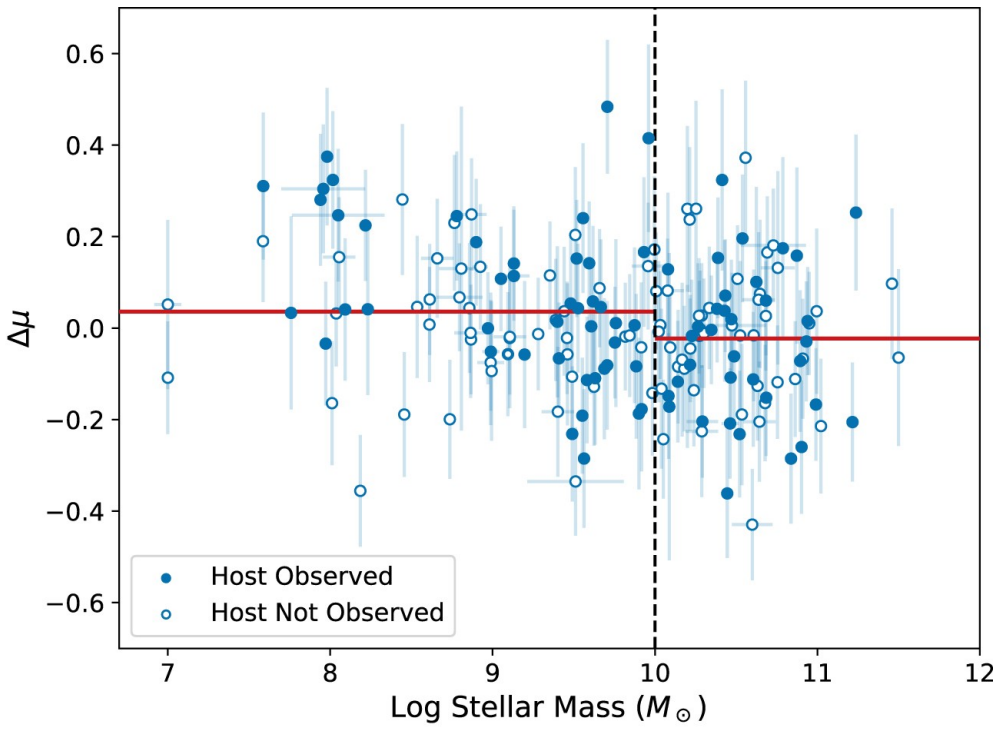


Figure 1: From Martin et al. (2024), showing the mass step in Hubble residuals is reproduced from the Foundation survey dataset that is used in Martin et al. (2024)'s analysis where the average Hubble residual is a step function once host galaxy stellar mass surpasses $10M_\odot$.

It was found that even after the corrections made using equation 1 to the SNe Ia light curves, that a correlation with a significance of 2.5σ with host galaxy stellar mass remained (Kelly et al., 2010; Lampeitl et al., 2010; Sullivan et al., 2010). This remaining correlation showed that SNe Ia are 10 per cent brighter in more massive host galaxies, which lead to a variance of the equation of state parameter for dark energy (w_Λ). Kelly et al. (2010), who found that SNe Ia in more massive host galaxies were$\simeq$ 0.1 mag brighter than SNe Ia in less massive host galaxies. Kelly et al. (2010) used 70 SNe Ia from the Sloan Digital Sky Survey with a redshift range $0.015 < z < 0.08$ and determined that when using μ_{obs} from 30 SNe Ia with host galaxies that have a stellar mass less than $10^{10.8}\mathrm{M}_\odot$ the equation of state parameter is $w_\Lambda = -0.78^{+0.152}_{-0.108}$ whereas for 30 SNe Ia with host galaxies with a stellar mass greater than $10^{10.8}\mathrm{M}_\odot$,$w_\Lambda = -1.03^{+0.217}_{-0.143}$.

For the standard ΛCDM model of the Universe, the density of dark energy is taken as constant which implies an equation of state parameter $w_\Lambda = -1$. And as can be seen in Kelly et al. (2010), using SNe Ia that are in less massive galaxies don't reflect the $w = -1$ that is accepted in the modern model of the Universe. The error goes beyond the uncertainty of the determination of w_Λ, showing that there is a systematic error in the standardisation of SNe Ia samples that are used in cosmology. This finding by Kelly et al. (2010) was corroborated by Lampeitl et al. (2010) who used 361 SNe Ia from the SDSS-II Supernova Survey and limited the redshifts of the sample to $z < 0.21$. Lampeitl et al. (2010) found the more massive and lesser star formation rate (SFR) galaxies, hosted SNe Ia that were $\simeq 0.1$ mag brighter than SNe Ia that were in galaxies of lower stellar masses. Finally, Sullivan et al. (2010) using 195 SNe Ia from the Supernova Legacy Survey within redshift $z < 0.85$ also independently found that SNe Ia in higher mass host galaxies are 0.06–0.09 mag brighter than SNe Ia in galaxies with lower stellar mass.

Although older, Kelly et al. (2010), Lampeitl et al. (2010), and Sullivan et al. (2010) independently found the correlation of Hubble residuals with the stellar mass of the supernovas host galaxy and provided the initial mass step correction to further standardise supernova light curves. Kelly et al. (2010), Lampeitl et al. (2010), and Sullivan et al. (2010) also demonstrates how the mass step is found across 3 different surveys and a redshift up to 0.85. Showing that the mass step correction is applicable systematically in the light curve

measurements of SNe Ia used as standardisable candles in cosmology, although Kelly et al. (2010), Lampeitl et al. (2010), and Sullivan et al. (2010) originally found and corrected for the mass step, it is not explained how the host galaxies global stellar mass is connected to the brightness of SNe Ia.

2.2 Progenitor Channels

It is currently accepted that the star that goes supernova producing the characteristic SNe Ia light curve is a carbon-oxygen (C-O) white dwarf (Hoyle, 1960). Although white dwarfs don't undergo supernova during their evolution, thus requiring other progenitor channels to be explored(Antonini et al., 2026). The two popular progenitor channels for SNe Ia are the single degenerate scenario (SD) and the double degenerate scenario (DD). SD proposes that the before an SN Ia event occurs the system that was there before, consisted of a C-O white dwarf that was accreting material from a non-degenerate companion star in a binary orbit(Whelan & Iben, 1973). Whilst the DD scenario presents the idea that the merger of two degenerate stars such as two C-O white dwarfs could also result in a supernova(Iben & Tutukov, 1984).

The SD and DD progenitor models have been popular for decades and with recent research indicating that there is more nuance to these two progenitor channels that can cause the variety of the light curves observed by astronomers. Deckers et al. (2022) study shows that out of 115 SNe Ia events that were detected by the Zwicky Transient Facility (ZTF), 67 per cent of the observed supernova light curves could effectively be recreated by the TURTLS model where the luminosity of the explosions are a direct result of Chandrasekhar-mass explosions with extended distributions and 0.4 to 0.6 $M_{\odot}$ of ^{56}Ni. Deckers et al. (2022) shows that over two thirds of SNe Ia could be resulting from Chandrasekhar mass C-O white dwarf explosions, whilst also demonstrating that 11 per cent of the SNe Ia light curves in the sample aren't reproduced from Chandrasekhar mass explosions and 19 per cent of the sample cannot be constrained using the varied masses and distributions of ^{56}Ni.

Deckers et al. (2022) also demonstrates a method to determine flux excesses in 6 light curves of the SNe Ia in the sample, Deckers et al. (2022) finds that these flux excesses are generally well produced by the non degenerate companion interaction models as well as the

^{56}Ni clumping models that also produced the light curves of the 67 per cent of the SNe Ia. Due to the limited sample size Deckers et al. (2022) concludes that SNe Ia likely have numerous pathways to explosion hence the varied observed light curves, but it appears that the majority can be modelled through the SD scenario. Deckers et al. (2022) urges more high cadence UV and NUV surveys with spectroscopic follow up be done to conclude with more confidence the explosion scenario of SNe Ia.

In a similar vein to Deckers et al. (2022), Burke et al. (2022) shows that using a variety of different models that from an unbiased sample of 9 SNe Ia that possess early light curve data 16 days before the epoch of peak flux. Are well produced by non-degenerate companion interaction with the ejecta, which supports the SD scenario interpretation. Burke et al. (2022) suggests that the ejecta from the mass accreting white dwarf, shock heats the surface of the non-degenerate companion star which produces the earlier mentioned flux excess in the light curves. This corroborates with the findings of the 6 early excess (EEx) SNe Ia in Deckers et al. (2022) which reported non-degenerate companion interactions effectively modelled the observed light curves.

Interestingly Burke et al. (2022) also reported that 8 of the 9 SNe Ia were near-UV to Blue in colour (i.e. SNe Ia that have negative colour parameter c), although Deckers et al. (2022) only reports 3 of the 6 having blue colour whilst the other three were red colour. This observed difference in colours is not reconciled through cross examination of either study. Both studies are still limited to small sample sizes which makes it difficult to make broader conclusions early time light curve behaviour and thus difficult to constrain the possible progenitor channels for SNe Ia. Regardless, both papers demonstrate the importance of high cadence near-UV surveys to detect more early light curves for SNe Ia.

Although light curve fitting and modelling has shown results in determining the possible progenitors systems of SNe Ia, there have been numerous studies on the correlations of SNe Ia light and the Host galaxy properties. Kelsey et al. (2022) finds through the analysis of 675 photometrically classified SNe Ia from the Dark Energy Survey 5-year sample, that blue SNe Ia are found in host galaxies that have lower stellar mass with higher star formation rates and are more blue in colour. Kelsey et al. (2022) makes the argument that the stars in these hosts are younger populations that are producing the more blue colours.

Although an older study, Anderson et al. (2015) finds a similar trend with SNe Ia light tracing out the blue light from host galaxies using pixel statistics method. Anderson et al. (2015) asserts that because the light from SNe Ia traces out the bluer region of host galaxies with lower SFR that progenitors systems that are less than 100 Myr old are extremely rare or do not exist. In comparison to Kelsey et al. (2022) who found that bluer SNe Ia correlate greater in galaxies with higher global star formation rates thus are more likely to be found in younger environments of the host galaxy. But Anderson et al. (2015) using the pixel method found that the more 'local' lower star formation rates host the bluer SNe Ia, leading to the older progenitor systems. The inconsistencies between Kelsey et al. (2022) andAnderson et al. (2015) can be explained through the correlations being found using global host galaxy properties and local host galaxy properties but this cannot be said conclusively without a encapsulating study of both datasets comparing the methods.

2.3 Early Flux Excess

Early flux excess in the light curves of SNe Ia are seen as bumps in the light curve before the flux of the SNe Ia has reached its peak. In 2012 a SN Ia was observed and the analysis of the light curve by Marion et al. (2016) showed an EEx flux in the V B and U band photometry. This was the first analysis of a 'normal' SNe Ia that had an EEx flux and the conclusion drawn was that the progenitor system likely was a non-degenerate companion, that was shock heated and was lucky enough to have the right viewing angle so that the excess could be observed.

Burke et al. (2022) uses high cadence data which was obtained from high cadence follow up observations using Las Cumbres Observatory and Swift, found 9 SNe Ia that had good early light curve data. Burke et al. (2022) found that the light curves were best explained via the shock heating of a non-degenerate companion where a C-O white dwarf was accreting mass from this companion and upon explosion, had ejecta interactions with the companion that results in the companion, at the right angle, emitting a near-U-V blue excess. This again corroborates with the discovery Marion et al. (2016) proposes that companion interactions with a non-degenerate star is a physical cause of the EEx observed.

Referring back to Deckers et al. (2022) who had a sample size of 6 EEx SNe Ia found

that the EEx colour and the lifetime of the flux excess is consistent with their modelling of both companion and circumstellar medium (CSM) interaction as well as ^{56}Ni clumping in the outer ejecta of the white dwarf. Deckers et al. (2022) argues that the EEx flux in SNe Ia is heterogeneous and can be caused by multiple different progenitors. Deckers et al. (2022) is to contrast the previous studies as they conclude, that to be more definitive regarding the mechanisms behind the EEx in flux, one day cadence surveys will enable fast spectroscopic follow up which will grant the ability to make more confident conclusions about the physics of the EEx.

Lim et al. (2023) is a case study on SN 2021hpr which exhibits the EEx in it's flux, similar to Marion et al. (2016) and Burke et al. (2022), the shock heating emission of the companion star for the white dwarf most suitably models the light curves of SN 2021hpr across the B, V, R and I bands from the Intensive Monitoring of Nearby Galaxies Survey which can be seen in Figure 2. Although the light curve fit well with a non degenerate companion model, Lim et al. (2023) could not find evidence of any stripped mass of the companion that would've occurred using spectroscopic follow-up. Lim et al. (2023) also acknowledge that they did not assess the viability of models that could cause the excess, such as double detonation models or disk originated matter models.

Lim et al. (2023) shows that the light curve deviates from a simple power law model as well as the companion model from Kasen (2010) and has a bump across all bands which can be seen in Figure 2. But the combination of both the power law and the Kasen (2010) companion models produce a curve that fits the observations.

The method used to detect the EEx fluxes in the SNe Ia data can vary, in Deckers et al. (2022) they use the residuals in the light curves that are found by subtracting the modelled light curve generated using the TURTLS model from the observed light curve. If a positive residual is found in the light curve within the calculated 6 days after the explosion epoch Deckers et al. (2022) classify that SNe Ia light curve as an EEx SNe Ia.

In Burke et al. (2022) the EEx flux SNe Ia are identified in a similar way to what was done in Lim et al. (2023) as Burke et al. (2022) uses the same companion shock cooling models present in Lim et al. (2023) from Kasen (2010). Although Lim et al. (2023) assumes SN2021hyp had a perfect viewing angle for the observed data, in Burke et al. (2022) the

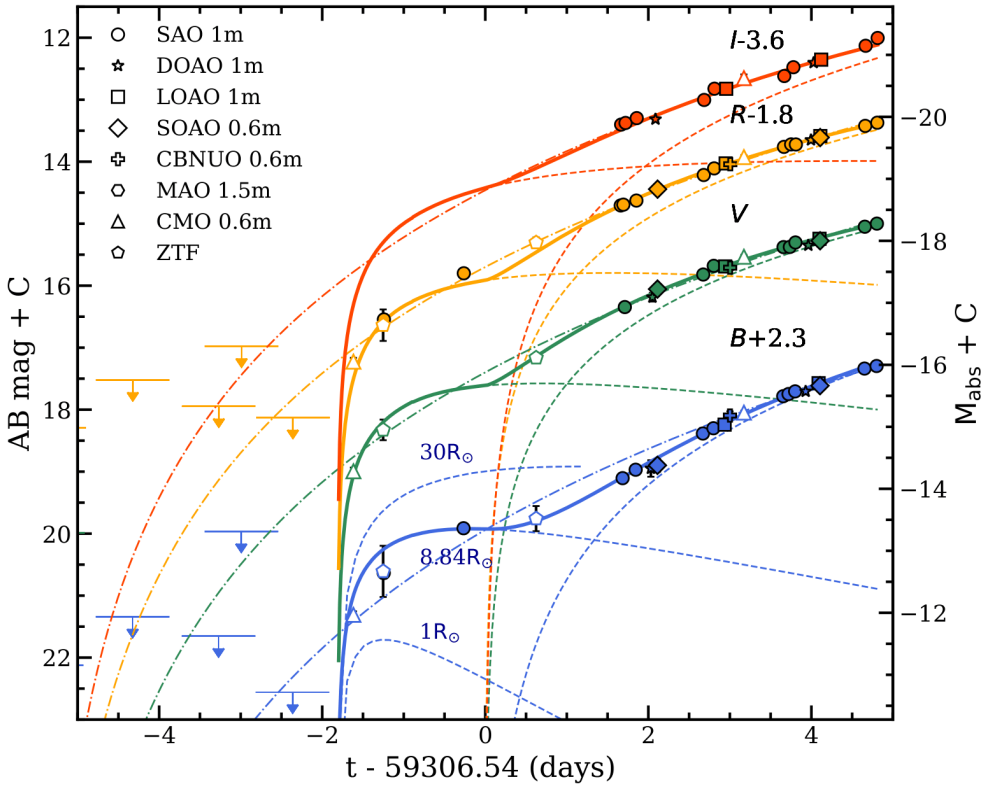


Figure 2: The light curves of SN2021hpr in magnitude space taken from Lim et al. (2023). The points are the observations, the dashed-dotted line is the light curve modelled by the simple power law model and the dashed lines are modelled light curves from Kasen (2010) which effectively model the light curves including the emission of light from the shock heating of a companion star. Finally, the solid lines are the two models combined and can be seen to line up well with the observations.

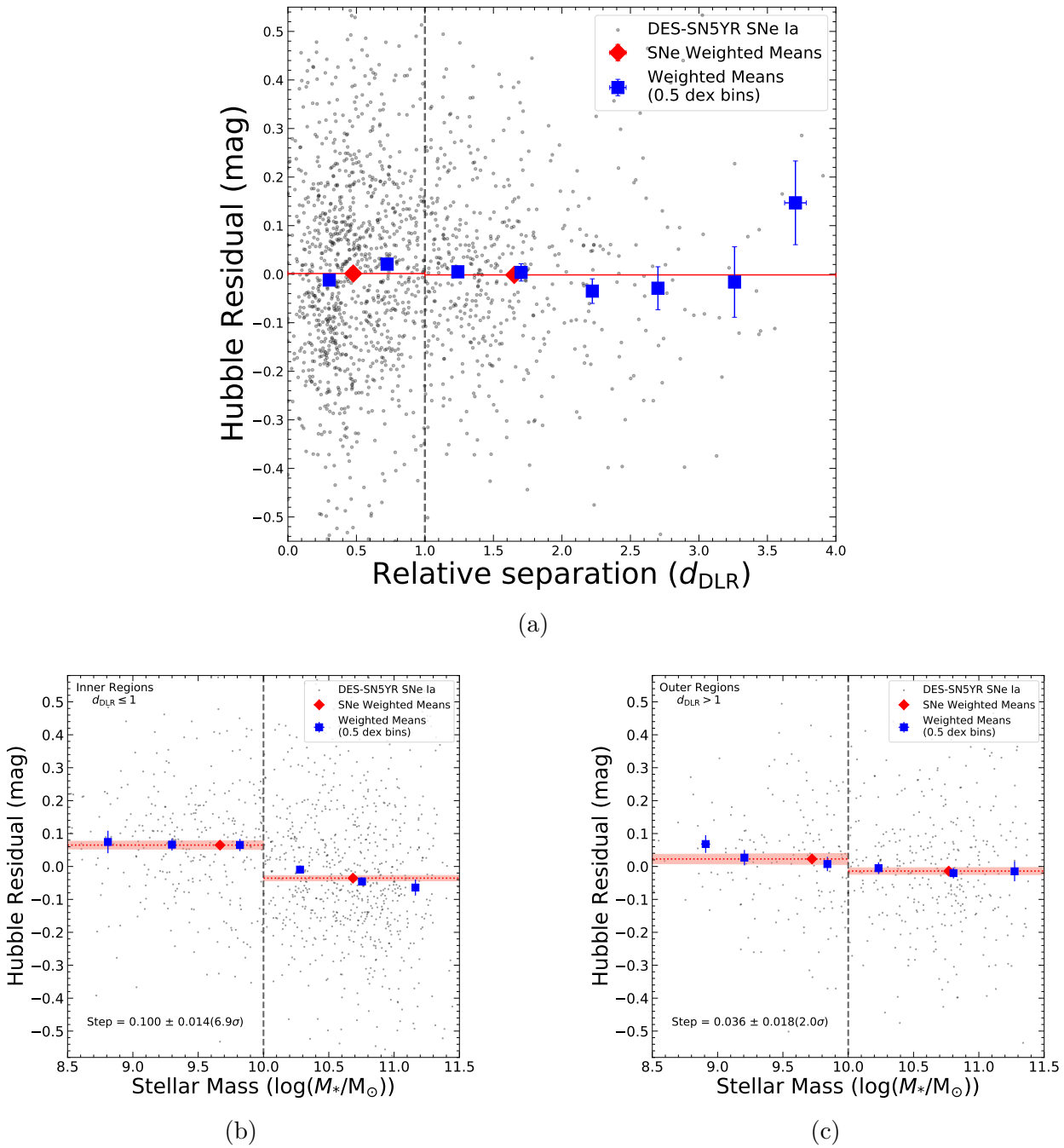
viewing angle is accounted for across the analysis of the nine early detected SNe Ia which leads to Burke et al. (2022) stipulating that they classify a EEx SNe Ia if there is data available within 5 days of first light, if the best fit viewing angle is less than 90 degrees and the residuals in the g band show a systematic decreasing from the earliest epochs to then matching the modelled light curve. Finally in Marion et al. (2016) the EEx flux SNe Ia was detected through the comparison of the observed light curve to the fitted power model light curve where the observed data had to be 3 times the measurement uncertainty.

3 Host Galaxy Environment and SN Ia diversity

3.1 Host Galaxy Properties

In chapter 2 it's clear that the environment of the SNe plays a factor in the shape of the light curve observed from an SN. Thus the properties of the host galaxies of SNe have been studied to determine the correlations between those properties and the locations of where SNe happen. Toy et al. (2025) uses a sample of 1533 SNe Ia from the 'five-year' sample from the DES SN program. Toy et al. (2025) using the Directional Light Radius method (DLR) to associate SNe Ia with a host galaxy and find that there is no step-like change in the Hubble residuals average across a normalised DLR, d_{DLR}, of 1. But when splitting the sample to 'inner regions' where $d_{DLR} < 1$ the Hubble residuals have a step in the averages of 0.100 ± 0.014 mag when split across the host galaxies with a stellar mass of 10. But in the outer regions ($d_{DLR} > 1$) Toy et al. (2025) finds that the step in the Hubble residuals when split by host's with stellar masses of 10 is 0.036 ± 0.018. Toy et al. (2025) makes the conclusion that SNe Ia in the outer regions are less affected by the global host galaxy properties, although Toy et al. (2025) notes that the different mechanics that produce this result is still unknown.

By searching for these correlations with the host galaxy properties, an intrinsic property of the SNe Ia's environment could explain this observation of SNe Ia in higher mass galaxies being brighter than those in lower mass galaxies. As seen in Toy et al. (2025), the outer regions of the host galaxy appear to be less affected by whatever is changing the observed



(a)

(b)

(c)

Figure 3: Taken from Toy et al. (2025), Figure a shows the mass step that can be seen in Figure 1 but as a function of the relative separation from the galaxy centre to the location of the SNe Ia. As a function of the relative separation, it's obvious that the step between Hubble residuals is greatly reduced. In images b and c which show the Hubble residuals for the inner regions and outer regions respectively, the mass step is found to be reduced for the SNe Ia in the outer regions of their host galaxies.

brightness of these SNe Ia.

Another interesting property to study is the global colour of the Host galaxy, Kelsey et al. (2022) finds that blue SNe Ia have higher homogeneity in the Hubble residuals. This higher homogeneity in blue SNe Ia was specifically for SNe Ia in a blue host galaxy environment with lower masses. Kelsey et al. (2022) found this homogeneity of blue SNe Ia by correcting for the global host galaxy colour ($U - R$) as a function of the colour of SNe Ia. Kelsey et al. (2022) interprets this stronger homogeneity post-standardisation as a sign that the older, more globally red galaxies have had more time for SNe to occur and create dust that reddens the SNe Ia. Which leads to another conclusion Kelsey et al. (2022) makes, which is that attenuating for dust doesn't reduce the scatter in the Hubble residual mass step to the point where it is fully accounts for the remaining scatter post-standardisation.

Another study, Ramaiya et al. (2025), finds correlations with SNe Ia and the stage of evolution that the host galaxy of the SNe Ia. Ramaiya et al. (2025) uses 1685 SNe Ia from the DES SN sample and for the host galaxies Ramaiya et al. (2025) uses a wide range of sources which include the Canada-France-Hawaii Telescope, the VLT Survey Telescope, the Subaru Hyper Suprime-Cam, the Visible and Infrared Survey Telescope for Astronomy and the Spitzer Space Telescope and the Herschel Space Observatory. This number of sources lets Ramaiya et al. (2025) cover a wide range of light emitted from host galaxies for a robust study. As can be seen in Figure 4, the colour nuisance parameter β changes based on the stage of evolution that the host galaxy is in. Ramaiya et al. (2025) shows that the stage of the host galaxy's evolution which changes properties such as mass and SFR also affects the luminosities of the SNe Ia in their host galaxies.

Connecting both Ramaiya et al. (2025) and Kelsey et al. (2022) shows that there is a clear affect of as galaxies age and progress in their evolutionary stages that affect the luminosity of the SNe Ia in those galaxies. In Ramaiya et al. (2025) β is noticeably different to the active star forming galaxies but once star formation ceases, the colour correction isn't as aggressive. Kelsey et al. (2022) as a low $U - R$ which is a proxy for a younger star forming galaxy finds that SNe Ia require a higher β which is what was found in Ramaiya et al. (2025). The same relation for more red galaxies, where $U - R$ is higher requires smaller β. This corroboration of an age-mass relation of host galaxies in Ramaiya et al. (2025) and Kelsey et al. (2022)

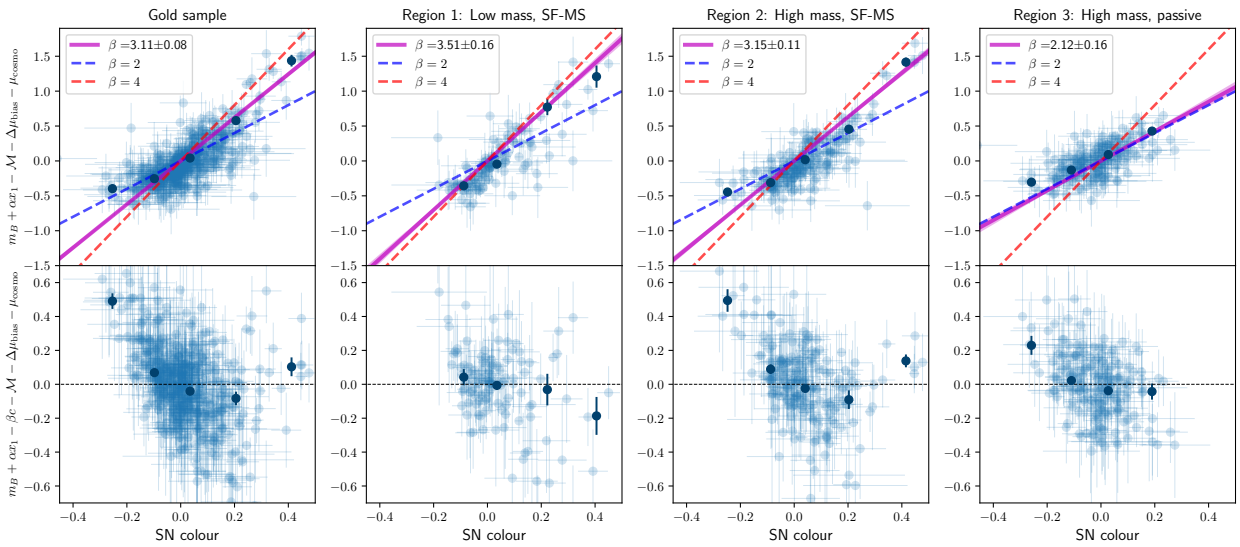


Figure 4: Taken from Ramaiya et al. (2025), the plots on top are the Hubble residuals plotted as a function of SNe Ia colour, without the colour calibration term in the Tripp equation. The bottom plots show the Hubble residuals whilst correcting for colour as a function of the SNe Ia colour. The gold sample is the sample of all the SNe Ia from the DES SN sample, whilst the remaining three are the plots of the gold sample divided by the evolutionary stage the host galaxy is in.

shows that the age and evolutionary stage of the host is a key relation in reducing the scatter in the Hubble residuals.

3.2 The Environment and Early Excess Flux

Host galaxy correlations are clearly important in the study of SNe Ia, as understanding the environment where SNe Ia are occurring in can help possibly explain some of the physics of the SN. We now understand that SNe Ia like to trace out the disk light of the host galaxy which and that SNe Ia luminosities are related to the age and stage of evolutionary of the host galaxy. Is this also true for the SNe Ia that present the 'bump' in the light curve?

Due to how small the samples sizes are for EEx SNe Ia, the studies of host galaxy properties correlating with properties of the SNe Ia is limited. Deckers et al. (2022) found, whilst acknowledging the small sample size doesn't allow them to make significant conclusions, that 5 of the 6 EEx SNe Ia have hosts with stellar masses lower then the mean of their sample $< \log(M_*/M_\odot) = 9.82$. Because of the connection to lower mass hosts, Deckers et al. (2022) postulates, stemming from Rigault et al. (2020) that the progenitor systems of EEx flux SNe

Ia are likely to be from a younger population.

A recent study by Wu et al. (2026) uses a sample of 16 SNe Ia found via detections of data 16 days before the B band maximum light. Wu et al. (2026) found 3 EEx Flux SNe Ia and could only associate two with a host galaxy. Conclusions are limited due to the limited data, regardless both the associated hosts of the 2 EEx SNe Ia had a lower stellar mass than the mean of the sample ($\log(M_*/M_\odot) = 9.94$). Whereas only one of the SNe Ia occurred in a host with a highest SFR of the sample ($3.64 M_\odot yr^{-1}$) whereas the second SNe Ia has the 4 lowest SFR of the sample.

It is safe to say that without more detections of EEx SNe Ia, correlations with host galaxy properties will not be able to reach a sample size that allows for statistically significant analysis. This results in a lack of conclusions being able to be drawn about the niche phenomena observed in these light curves.

4 Fink, Rubin and LSST

Over the course of this review it has become obvious that more and earlier data would provide the next steps for both progenitor channel and correlation constraining of SNe Ia with host properties. If more SNe Ia in hosts can be collected and much earlier before peak flux, then the correlations with host galaxy properties can be more easily studied which enables improvements for standardisation methods.

The EEx SNe Ia are difficult to find, with most studies not being population studies but rather case studies, and the few population studies result in a sample size of 6 and 3 (Burke et al., 2022; Deckers et al., 2022; Wu et al., 2026). The Vera C. Rubin Observatory which will be referred to as Rubin from here on out, is located in El Peñón peak of Cerro Pachón in Chile (Bianco et al., 2022). The observatory hosts a 8.36 metre aperture and will have filters across a wavelength range of 320.5 nanometres to 1084.5 nanometres (Bianco et al., 2022). The viewing cadence of LSST has an average revisit time of 3 days across the whole southern sky which, covers an area of 18000 deg^2 which gives the median number of revisits at 825 per site (Bianco et al., 2022; Ivezić et al., 2019). As can be seen in Figure 5 Some portions of the Southern Sky will be visited more than others but the main Wide-Fast-Deep

(WFD) survey will have an average of 900 revisits across the 10 years that LSST will operate over.

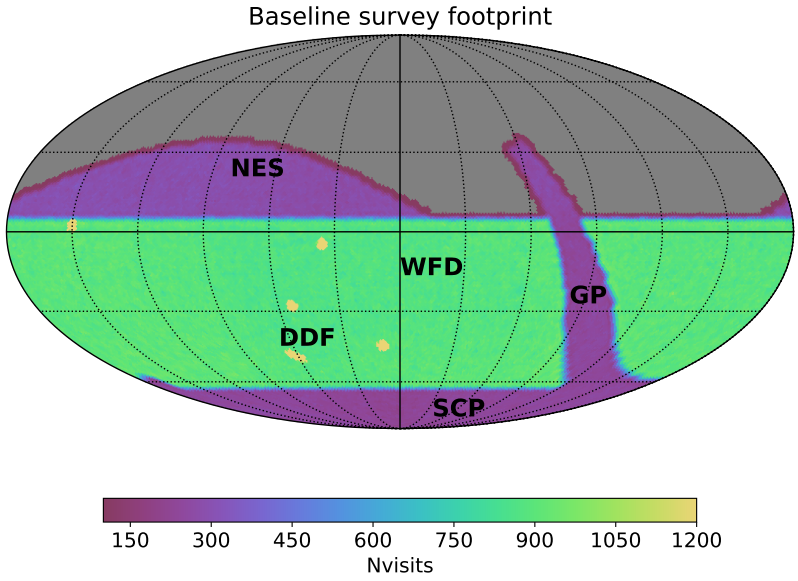


Figure 5: The baseline number of visits that Rubin will make across the Southern Sky. Taken from Bianco et al. (2022), the different areas on the map show the main survey, WFD, and additional mini surveys. With NES being the North Ecliptic Spur, DDF represents the Deep Drilling Fields, GP is the Galactic Plane and SCP is the South Celestial Pole. It is easy to see that different surveys using the Rubin Observatory require different viewing cadences to enable the collection of data that is desired.

Over the course of the survey the expected number of alerts generated each night will be 10 million per night and within 60 seconds, transients will be reported (Möller et al., 2021). Of these alerts it is expected that there will be $\approx$ 1100 SNe per night. This should enable the opportunity to discover more SNe Ia which will increase the odds of finding the EEx SNe Ia and could enable the beginnings of a population study with respect to their host galaxy properties.

To handle this number of alerts, we employ the use of the Fink broker which allows for the processing of the large number of alerts from the alert stream of Rubin. Fink does the ingestion of the data of the LSST alert stream, annotation of the alerts such as early light curve classification and is capable of customisable filtering of the alerts (Möller et al., 2021).

Fink employs machine learning methods of classification to disentangle the types of SNe that will be discovered by Rubin. These machine learning classifiers use an active learning

approach that can take a smaller sample of training data and continually retrain the classification models to improve the models as the survey progresses in time(Leoni et al., 2022). This means that SNe Ia will be readily classified from the LSST alert stream producing science-ready images in mere minutes.

5 Project Aims

Here we simply state the aims of the project,

- Curate a sample of Extra-galactic transients through Fink.
- Use the Fink broker to identify the Sn Ia in those transients using spectroscopic follow-up and machine learning algorithms.
- Use SALT3 to measure SN Ia properties c and x_1
- With a large enough sample, find EEx SNe Ia using the method laid out in Deckers et al. (2022)
- Find correlations of EEx SNe Ia with DLR, host galaxy mass, global colour and SFR.

6 Planned Methodology

The proposed method for this project will be to ingest data from Fink, filtering the incoming data based on the scores that the classifier Fink uses for SNe classification to build an initial sample of SNe. Following on from that initial sample the method will be reproducing the method followed by Deckers et al. (2022). Who made cuts based on colour parameters derived from the application of the SALT2 model, where SNe Ia with $-0.3 \leq c \leq 0.3$ are cut from the sample. Cuts are then made based on the stretch parameter $-3.0 \leq x_1 \leq 3.0$, then a cut based on the peak magnitude in the green band of the observation $-19.95 \leq peak_g \leq -18.35$ and finally the first detection of the SNe Ia needs to be 14 days before the peak in the green bands luminosity was reached. Deckers et al. (2022) argues that this is to remove perculier SNe Ia from the data set and leaves only 'normal' SNe Ia in the dataset for analysis.

It is unclear whether Deckers et al. (2022) does the following method for detection of EEx SNe Ia on the original starting sample of 127 SNe Ia or if it is on the sample of 115 SNe Ia post quality cuts. It is a reasonable assumption that the detection of EEx SNe Ia was done on the sample post quality cuts.

The method to detect the EEx SNe Ia used in Deckers et al. (2022) assess residuals in the light curves in flux space that are found by taking the difference between the best fit model light curve from the SALT2 model and the observed light curve. Specifically focusing on the span of time 6 days after the estimated explosion epoch which is determined from the best fit model. Although for this project, the SALT3 light curve model will be used due to its improved performance in the UV and blue wavelengths when compared to the SALT2 model. This is because these bumps, as mentioned in chapter 2 subsection 2.3, are universally near UV to blue throughout the excess.

Following the identification and classification of the EEx SNe Ia, the assigning of host galaxies will be done using the Directional Light Radius method, following this assignment the correlations of the EEx SNe Ia population in the sample with the properties of the host galaxies will be determined. These properties include but are not limited to the sSFR, stellar mass of the galaxy and the global colour of the galaxy. The colour of the EEx SNe Ia will also be apart of these correlative analyses to determine any correlations with host galaxy properties and the colour of the EEx SNe Ia.

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